

Camelyon17

Canine cSCC

MIDOG 2021

MIDOG++

	H0	H1	H2	H3	H4	cs2	p1000	nz20	nz210	g450	XR	S360	CS2	MCT (c)	lung (c)	lymph. (c)	STS (c)	breast (h)	melan. (h)	NET (h)
ERM	.972	.943	.895	.988	.977	.862	.534	.760	.905	.866	.879	.847	.855	.911	.886	.870	.927	.893	.910	.881
GroupDRO	.952	.951	.893	.985	.982	.850	.606	.785	.902	.870	.879	.839	.826	.910	.882	.879	.923	.838	.916	.878
IRM	.976	.935	.902	.819	.981	.853	.559	.774	.835	.860	.872	.840	.837	.917	.881	.862	.925	.638	.918	.887
CORAL	.978	.957	.923	.989	.983	.853	.702	.772	.918	.869	.879	.820	.840	.915	.886	.874	.919	.897	.919	.883
MixStyle	.981	.904	.792	.986	.956	.880	.632	.806	.890	.853	.882	.865	.846	.918	.881	.840	.919	.891	.918	.869
Fish	.974	.838	.880	.977	.887	.807	.792	.797	.940	.884	.874	.820	.840	.899	.873	.775	.903	.880	.909	.855
LISA	.977	.929	.886	.983	.963	.855	.734	.784	.890	.879	.875	.847	.860	.908	.872	.862	.925	.874	.911	.862
Macenko†	.982	.915	.951	.984	.976	.920	.819	.927	.909	.827	.889	.867	.860	.921	.893	.898	.938	.905	.914	.873
Macenko (per tile)	.989	.938	.985	.984	.557	.885	.804	.900	.897	.783	.885	.871	.846	.910	.879	.877	.925	.897	.900	.877
H&E jitter	.988	.958	.977	.987	.985	.916	.850	.936	.938	.875	.903	.888	.875	.916	.882	.880	.932	.909	.916	.884
TIA-style†	.987	.934	.978	.990	.960	.919	.823	.931	.923	.842	.899	.874	.871	.918	.886	.890	.946	.910	.911	.877
SimCLR	.986	.961	.929	.987	.979	.864	.607	.834	.900	.875	.877	.836	.853	.915	.884	.884	.923	.909	.913	.882
SAM	.974	.963	.900	.986	.979	.861	.624	.802	.926	.877	.904	.861	.851	.920	.892	.886	.935	.912	.921	.884
BN-adapt (ERM)†	.990	.929	.970	.974	.987	.903	.882	.930	.928	.883	.890	.886	.870	.916	.886	.868	.909	.888	.905	.879
Lunit BT FT	.966	.954	.895	.986	.970	.923	.853	.833	.937	.876	.898	.877	.878	.903	.877	.872	.922	.908	.892	.863
Lunit BT FT + H&E jitter	.959	.947	.902	.988	.988	.937	.888	.937	.936	.866	.899	.888	.885	.900	.874	.883	.924	.912	.913	.868
Lunit DINO FT	.993	.988	.986	.992	.986	.936	.859	.959	.965	.901	.882	.850	.829	.912	.862	.896	.916	.899	.921	.881
Lunit DINO FT + H&E jitter	.993	.982	.995	.994	.991	.935	.871	.966	.969	.902	.883	.883	.858	.917	.867	.904	.923	.903	.925	.887
RN50-IN probe	.969	.906	.948	.950	.964	.854	.772	.838	.847	.789	.623	.674	.644	.620	.581	.571	.658	.629	.624	.613
Lunit BT probe	.992	.969	.965	.984	.977	.880	.879	.935	.919	.870	.813	.797	.790	.801	.757	.739	.844	.828	.838	.811
Lunit DINO probe	.992	.975	.987	.989	.952	.943	.890	.956	.956	.910	.751	.752	.736	.768	.713	.712	.788	.778	.800	.731

Outlined: worst domain of the arm (rows with every domain present).

■ Reference ■ Self-supervision ■ Test-time adaptation
 ■ DG objectives ■ Sharpness ■ Pretrained representations
 ■ Stain & color

0.5 0.6 0.7 0.8 0.9 1.0
test AUROC (seed mean, ID-val selection)

† Transductive: uses the unlabeled images of the test domain at prediction time.